Does Female Leadership Relate to Productivity-Wage Gaps in Colombia?

Applied Econometrics ECON 6375 Paper

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Introduction

- Across the world, workers are paid substantially less than the value they produce. For example, estimates for Colombia suggest workers produce approximately 40% more than they earn.
- Microeconomic theory explains that the productivity-wage gap arises under imperfect competition where firms have wage-setting power.
- Yet this gap is not uniform: management structure shapes how aggressively firms exploit their wage-setting power.
- In particular, female leaders place greater weight on worker welfare and are less likely to engage in anticompetitive behavior.

This paper examines whether firms with a larger share of female leaders are associated with a lower productivity-wage gap.

Data

- Firm-level data from the Annual Manufacturing Survey (Encuesta Anual Manufacturera, EAM) of Colombia.
- Unbalanced panel of 11,235 firms between 2000-2019.
- Manufacturing sectors in developing economies are dominated by small and medium-sized enterprises.

Table: Descriptive Statistics

Variable	Mean	SD	Min	Median	Max
Share of female executives (%)	45.192	39.925	0.000	49.812	100.000
Firm's share of industry employment (%)	0.020	0.066	0.000	0.004	1.000
Share of workers in production (%)	61.951	21.743	0.000	66.667	100.000
Gross output ($ millions USD)	1.219	536.010	756.959	11.356	2,968.214
Firm age (years)	4.262	3.422	1	3	20
Number of workers	56	146	1	20	4921
Wage per worker ($ thousands USD)	1.827	91	349.180	2.296	1,498.014

Productivity-Wage Gap Definition

Following Yeh et al. (2022) we obtain the firm's markdown (ν) as:

$$\nu \equiv \frac{R'(L)}{w(L)} = \varepsilon_L^{-1} + 1 \tag{1}$$

After some algebraic manipulation using the production approach by De Locker et al. (2020):

$$\nu = \frac{\theta^L}{\theta^M} \cdot \frac{p^m M}{w(L)L}. \tag{2}$$

Where θ^L and θ^M are the input-output elasticities of labor and materials.

Markdown estimation - Production Approach

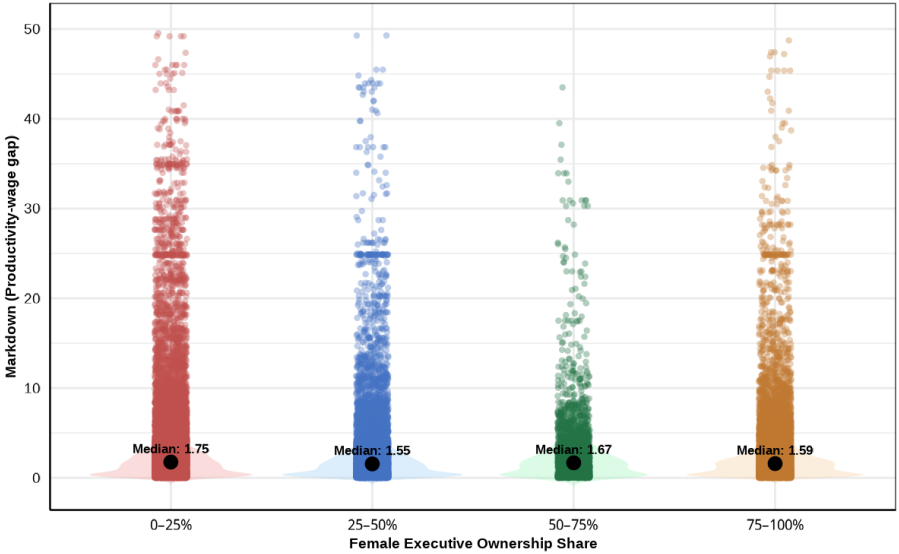
Translog production function GMM estimation:

$$\begin{aligned}\log(GO_{it}) = \beta_0 &+ \beta_1 \log(K_{it}) + \beta_2 \log(L_{it}) + \beta_3 \log(M_{it}) + \beta_4 (\log K_{it})^2 + \beta_5 (\log L_{it})^2 \\ &+ \beta_6 (\log M_{it})^2 + \beta_7 \log(K_{it}) \log(L_{it}) + \beta_8 \log(K_{it}) \log(M_{it}) + \beta_9 \log(L_{it}) \log(M_{it}) \\ &+ \gamma_t + \varepsilon_{it}.\end{aligned} \tag{3}$$

As follows:

$$\begin{aligned}\hat{\theta}_{L,it} &= \hat{\beta}_2 + 2\hat{\beta}_5 \log(L_{it}) + \hat{\beta}_7 \log(K_{it}) + \hat{\beta}_9 \log(M_{it}), \\ \hat{\theta}_{M,it} &= \hat{\beta}_3 + 2\hat{\beta}_6 \log(M_{it}) + \hat{\beta}_8 \log(K_{it}) + \hat{\beta}_9 \log(L_{it}).\end{aligned} \tag{4}$$

Markdown and Female Leadership



Empirical Framework

Linear regression model:

$$\log(\text{Markdown}_{it}) = \alpha + \beta\,\text{FemaleLeadership}_{it} + \boldsymbol{\gamma}'\mathbf{X}_{it} + \alpha_t + \alpha_s + \alpha_r + \varepsilon_{it} \tag{5}$$

where:

- $\log(\text{Markdown}_{it})$ is the logarithm of the productivity-wage gap for firm i in year t
- $\text{FemaleLeadership}_{it}$ denotes the share of women in executive and ownership positions
- The vector $\mathbf{X}_{it}$ includes a rich set of firm-level controls designed to isolate the leadership channel from confounding variation in productivity, scale, and workforce composition.
- α_t, α_s, and α_r are year, industry, and region fixed effects.
- ε_{it} is the error term.

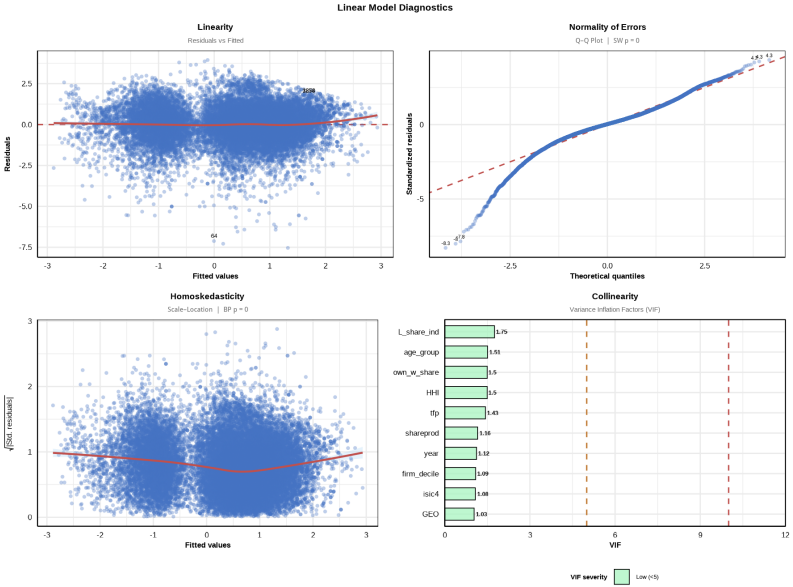
Regression Results

Table: Regression Results - Dependent variable: log(Markdown)

	(1)	(2)	(3)
Share of Female Executives	-0.0010***	-0.0010***	-0.0011***
	(0.0001)	(0.0002)	(0.0002)
Firm's share of employment in the industry		2.0356***	2.4268***
		(0.3723)	(0.3713)
Share of production workers		-0.0021***	-0.0018***
		(0.0003)	(0.0003)
Industry HHI		0.5729***	0.5719***
		(0.0634)	(0.0635)
TFP		0.1557***	0.1578***
		(0.0039)	(0.0039)
Additional controls		Size & Age	Size & Age
FE: Year	X	X	X
FE: Region	X		X
FE: Industry	X		
N	48,292	31,664	31,664
R^2	0.5097	0.1243	0.1386

Note: $^{***}p < 0.001$; $^{**}p < 0.01$; $^{*}p < 0.05$; $^{\cdot}p < 0.1$

Gauss-Markov Multiple Regression Assumptions



Conclusion

- Management structure is a meaningful margin of variation in firms' wage-setting behavior.
- A larger share of female executives is associated with less aggressive exploitation of monopsony rents.
- This is consistent with evidence that female leaders place greater weight on worker welfare and interpret productivity signals differently from their male counterparts.
- Policies promoting gender diversity in executive and ownership positions (board gender quotas, bias-free promotion practices, and mentorship pipelines) can increase worker welfare, reduce monopsony power and increase overall economic efficiency.